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Kim et al.

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(54) **FIXING DEVICE FOR INSTALLING PRESSURE-TYPE AIRBAG SENSOR**

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B60R 21/013 (2006.01)
B60R 21/00 (2006.01)

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CPC ***B60R 21/013*** (2013.01); ***G01L 5/0052*** (2013.01); *B60R 2021/0006* (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — John E Breene
Assistant Examiner — Nigel H Plumb
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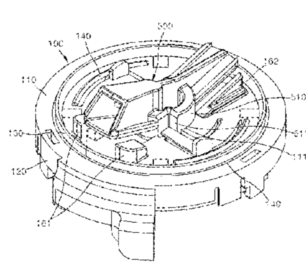
(57) **ABSTRACT**

The present invention provides a fixing device for installing a pressure-type air bag sensor, the fixing device comprising: a disc-shaped holder; a cover coupled to the holder and having a sensor part disposed on the top thereof; and a connector housing disposed on the bottom of the cover, wherein the holder has a holder body formed in a disc shape, an outer wall having a predetermined height is formed along the outer circumference of the holder body, a ring-shaped seal is formed through insert-injection molding to have a predetermined radius inside the holder body, and protrusion parts having a predetermined curvature are formed at the upper surface part and the lower surface part of the holder body.

5 Claims, 18 Drawing Sheets



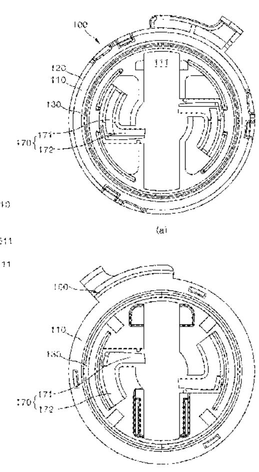


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FIG. 1

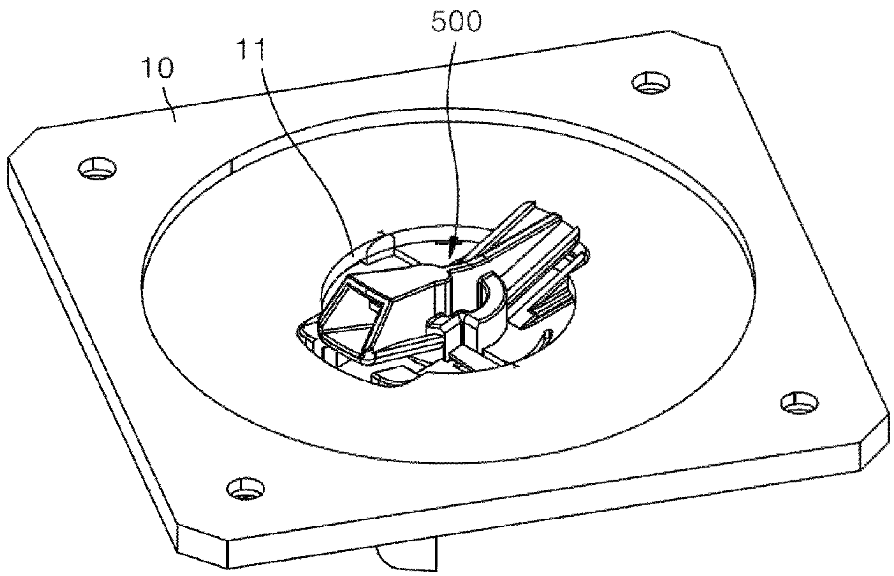
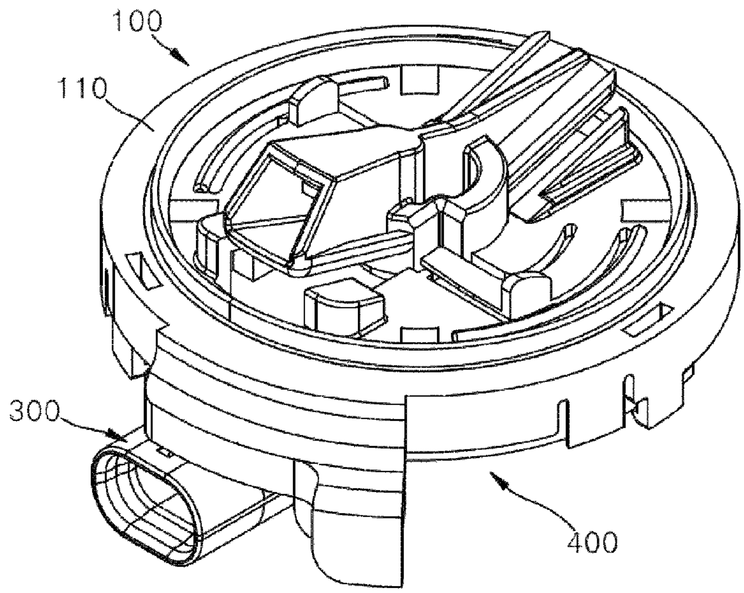
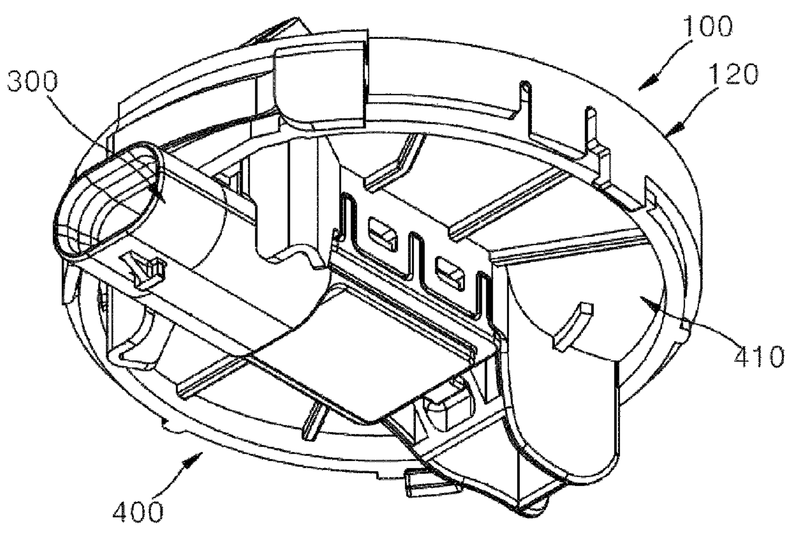


FIG. 2



(a)



(b)

FIG. 3

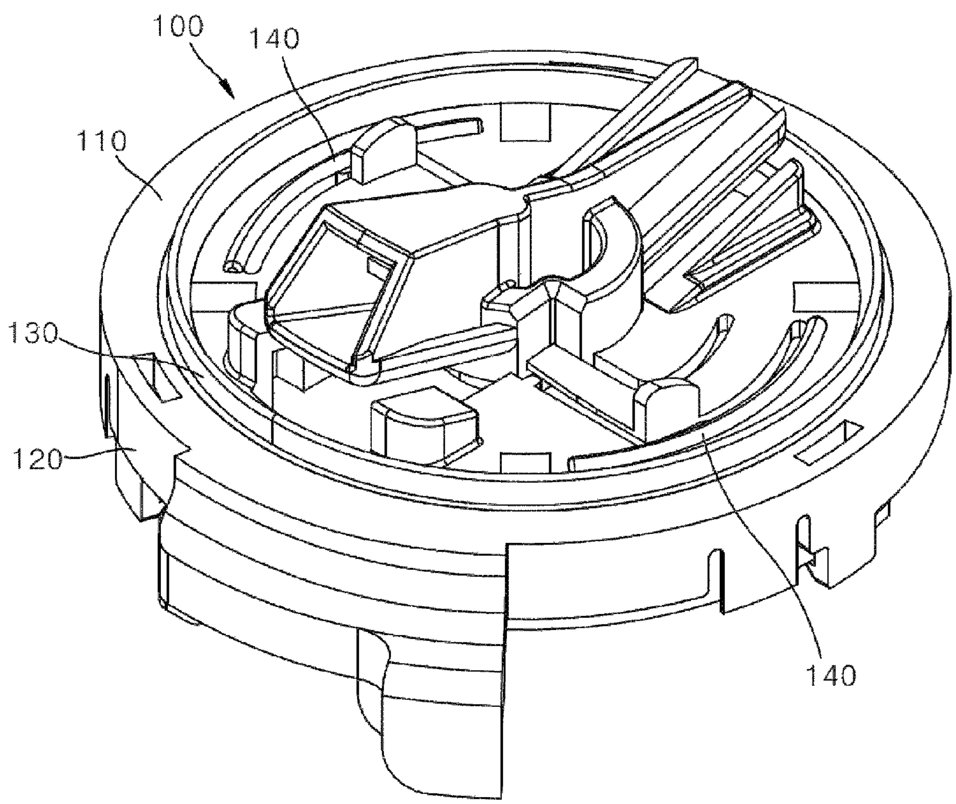


FIG. 4

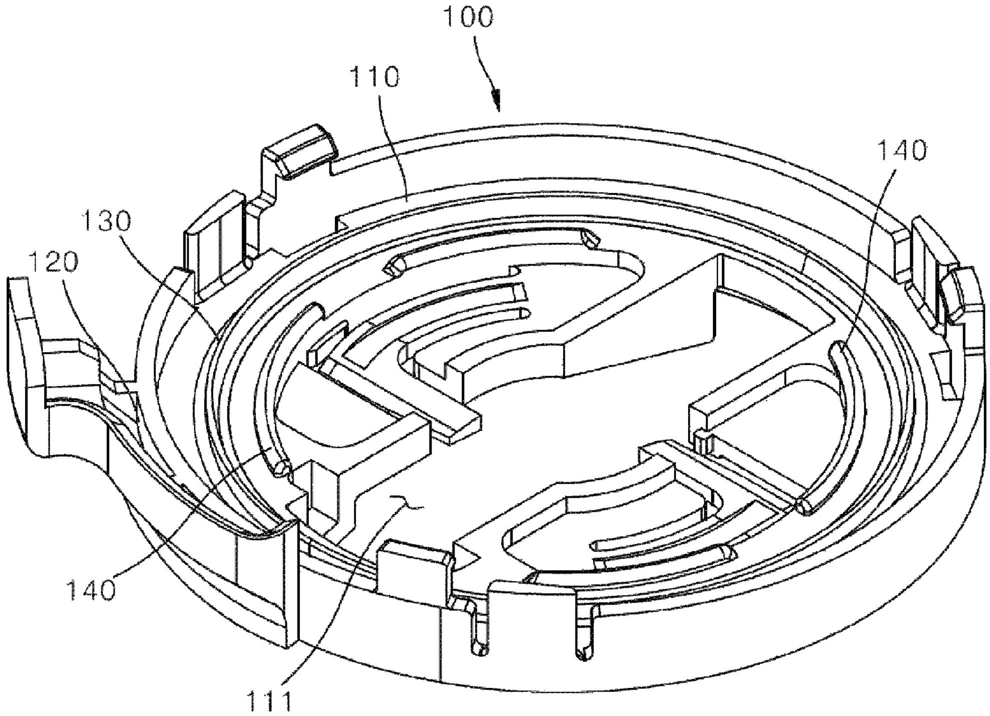


FIG. 5A

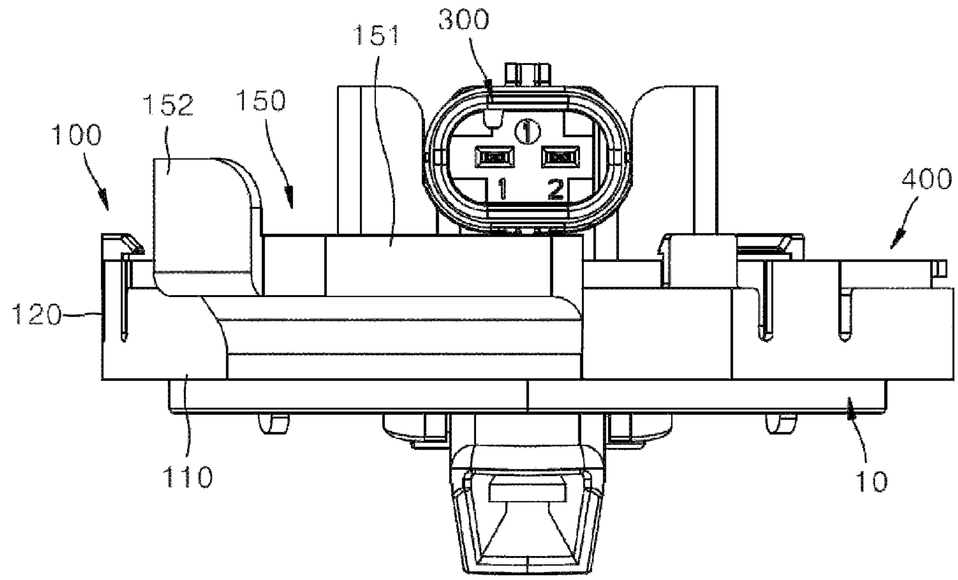


FIG. 5B

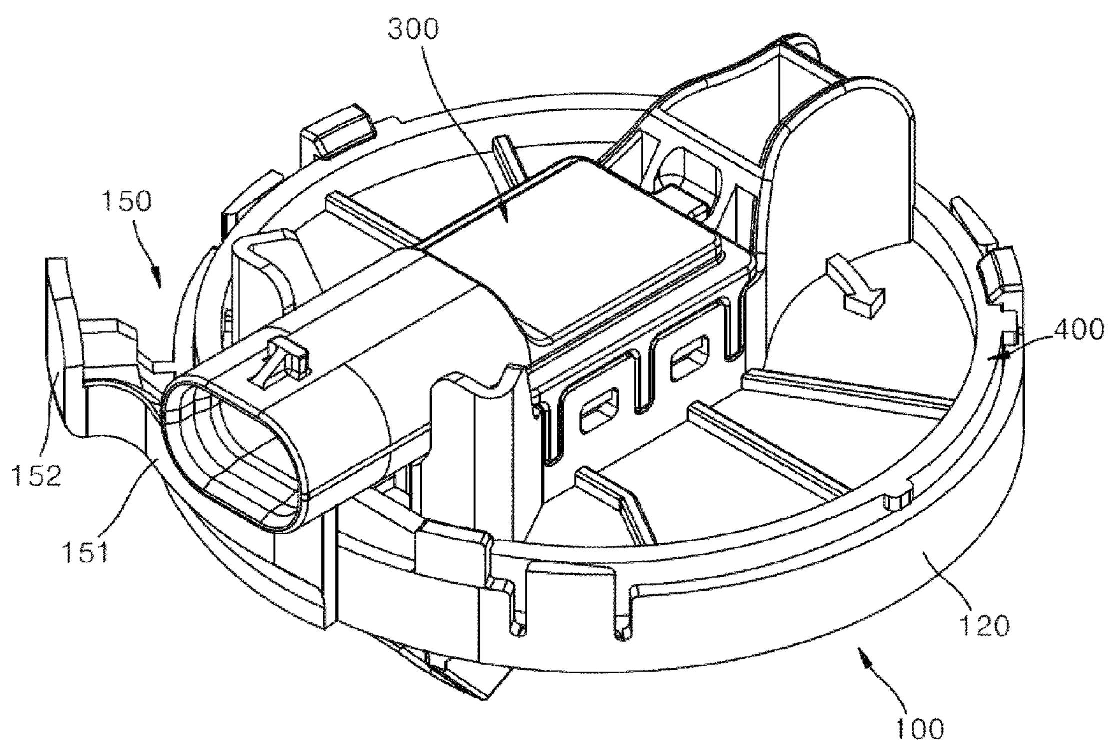


FIG. 6

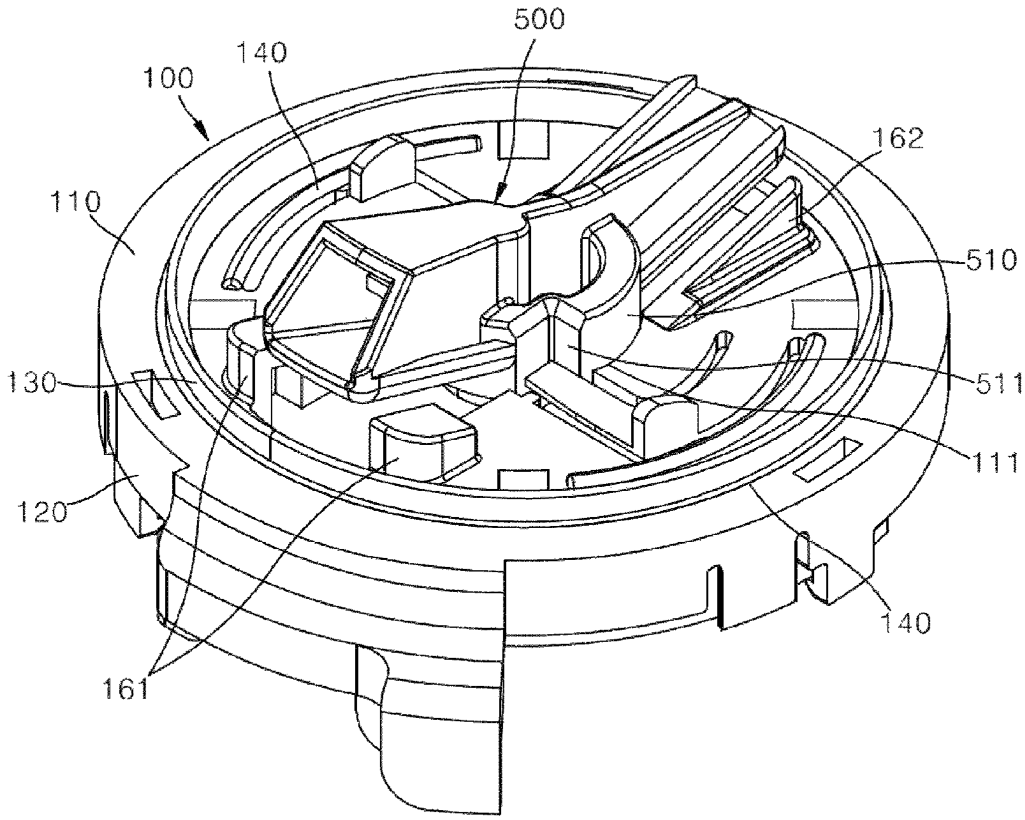


FIG.7

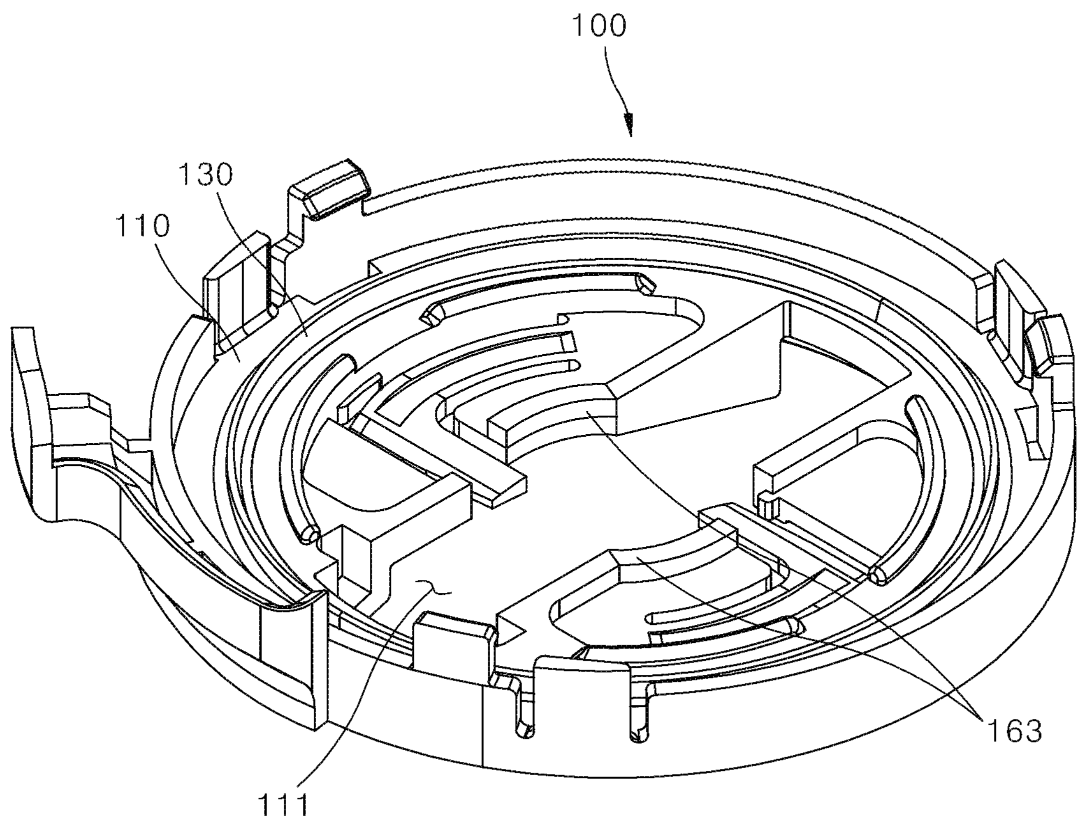
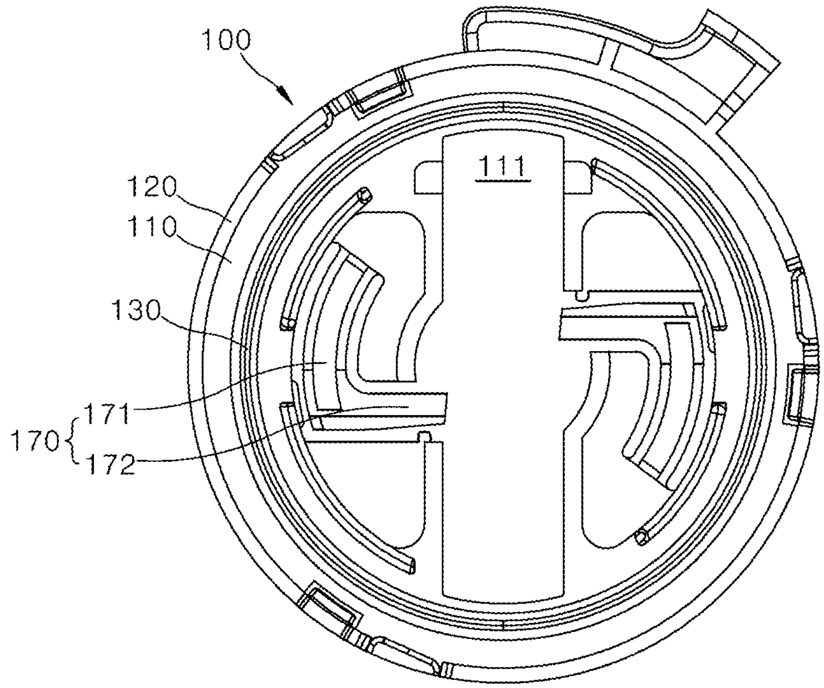
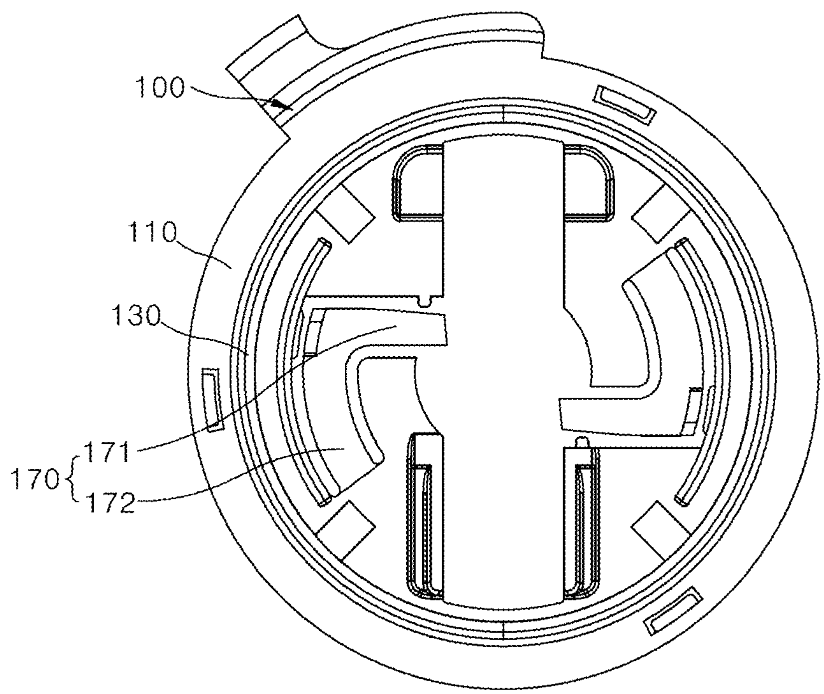


FIG.8



(a)



(b)

FIG.9

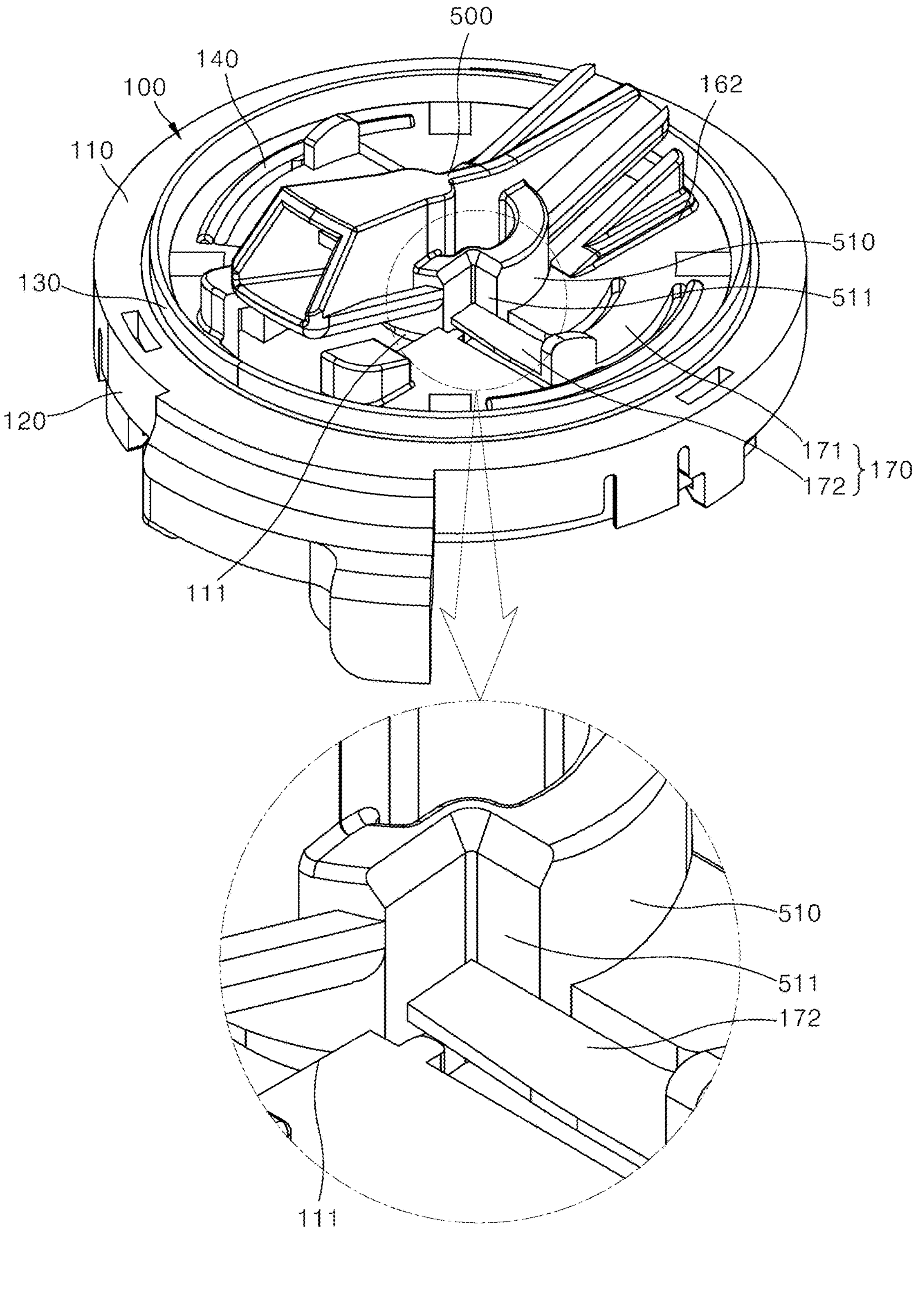

500
140
100
110
162
510
130
511
120
171
172
170
111
510
511
172
111

FIG.10

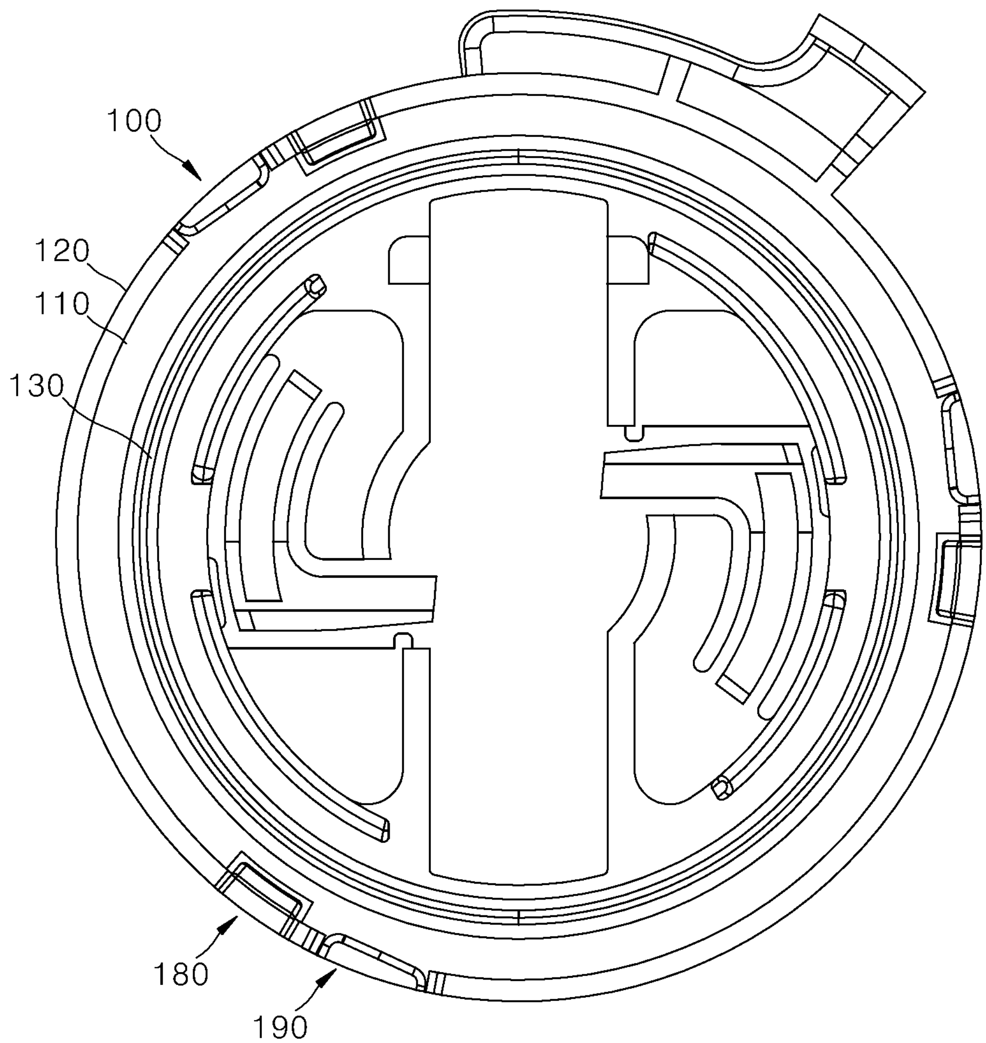


FIG. 11A

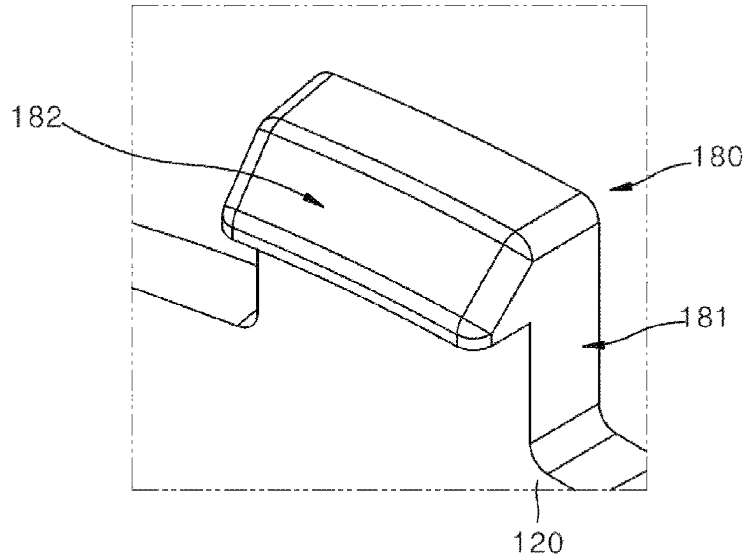


FIG. 11B

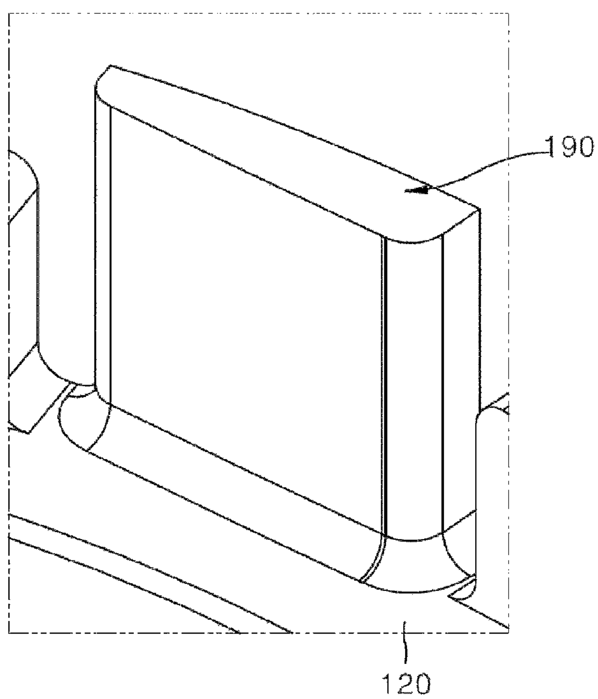


FIG. 12

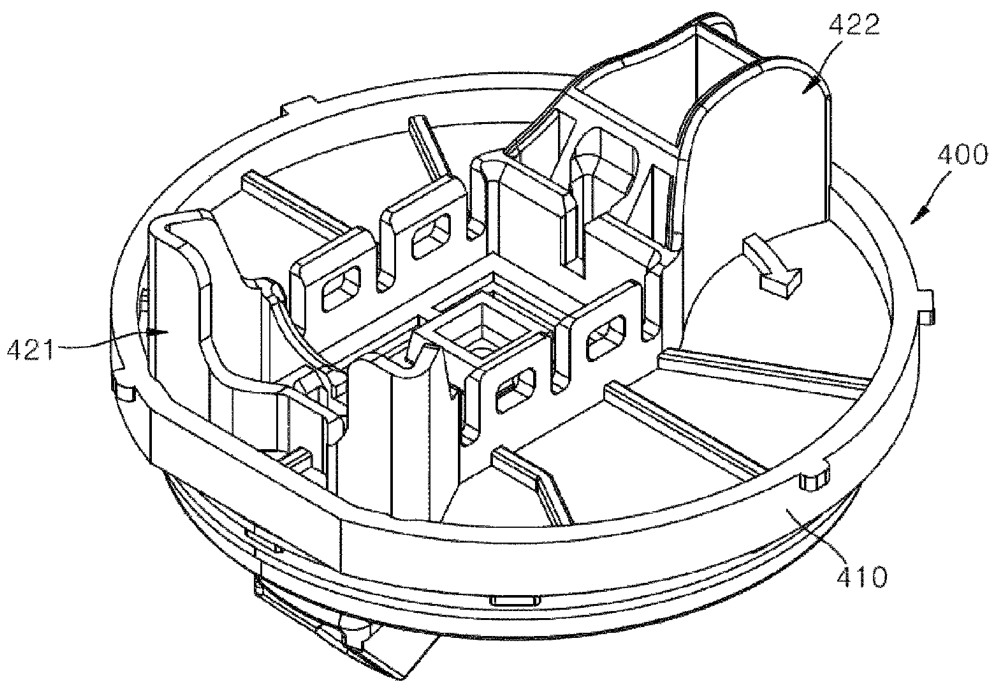


FIG. 13

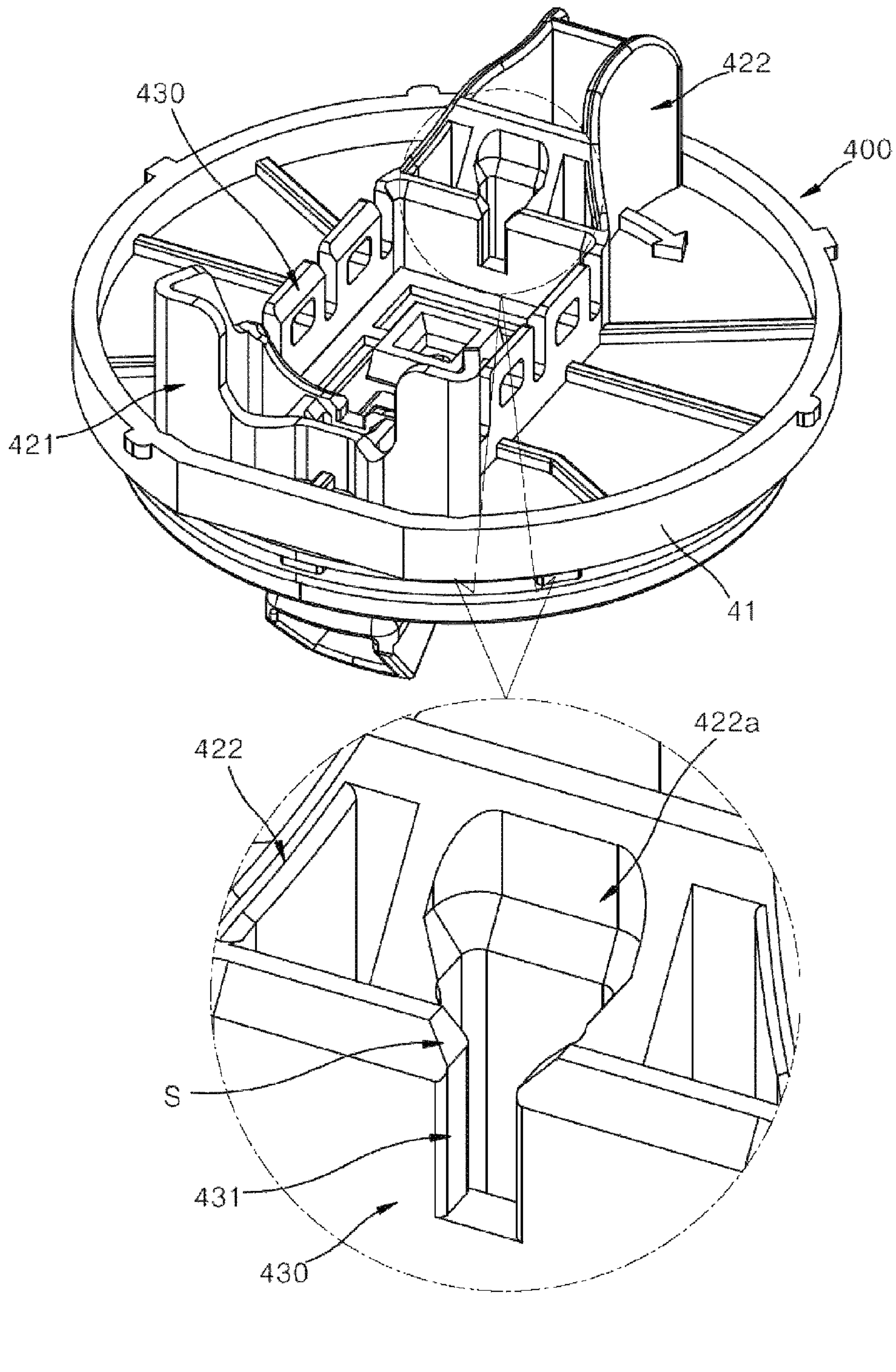


FIG. 14

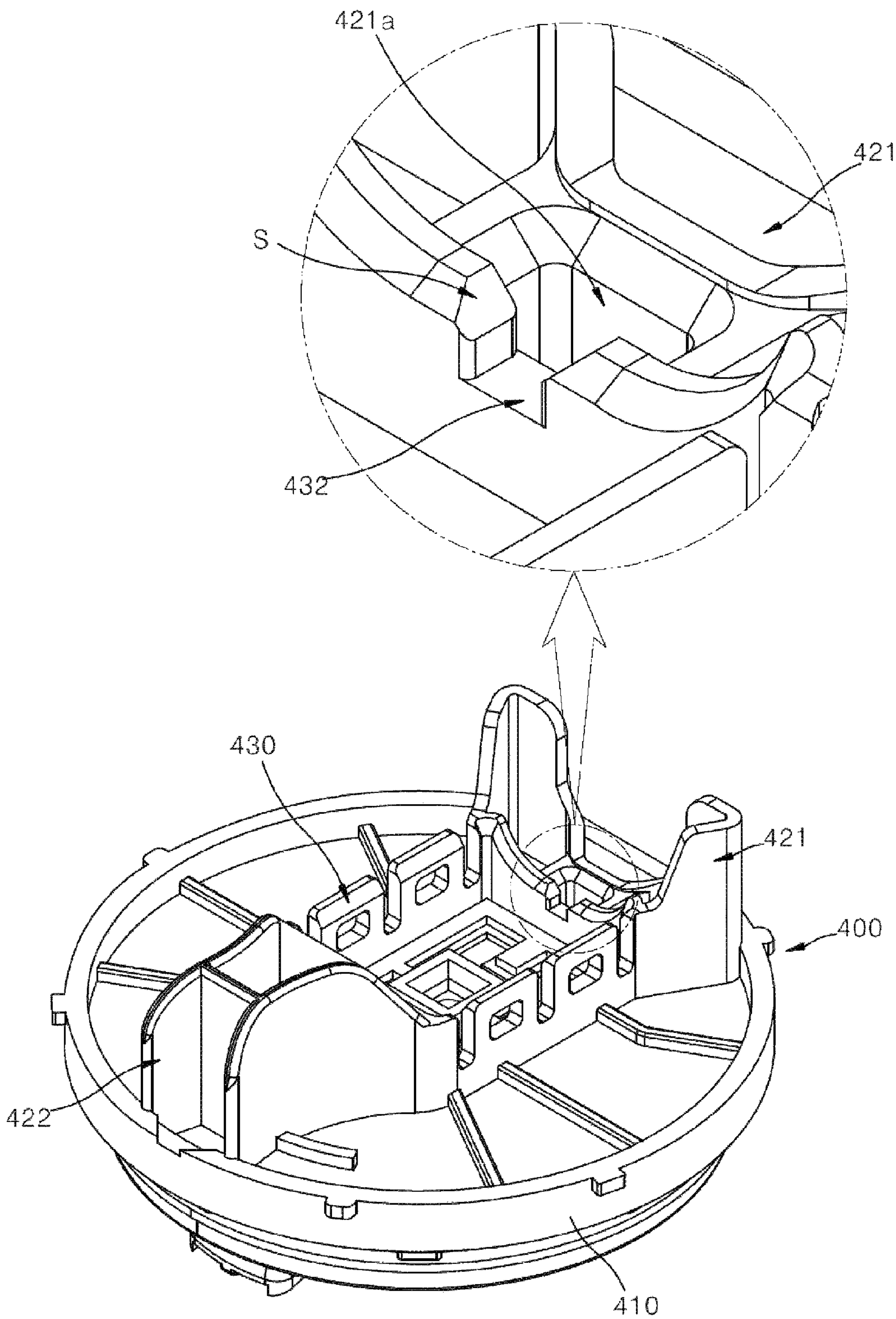


FIG. 15

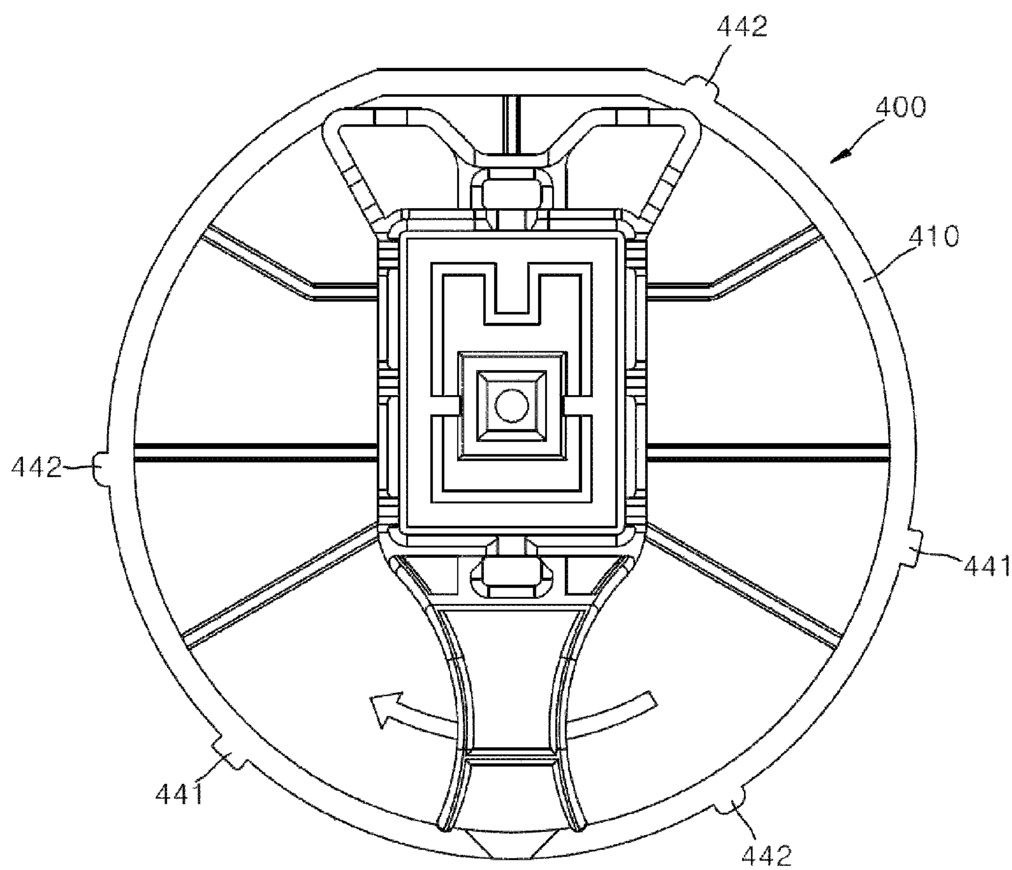


FIG. 16

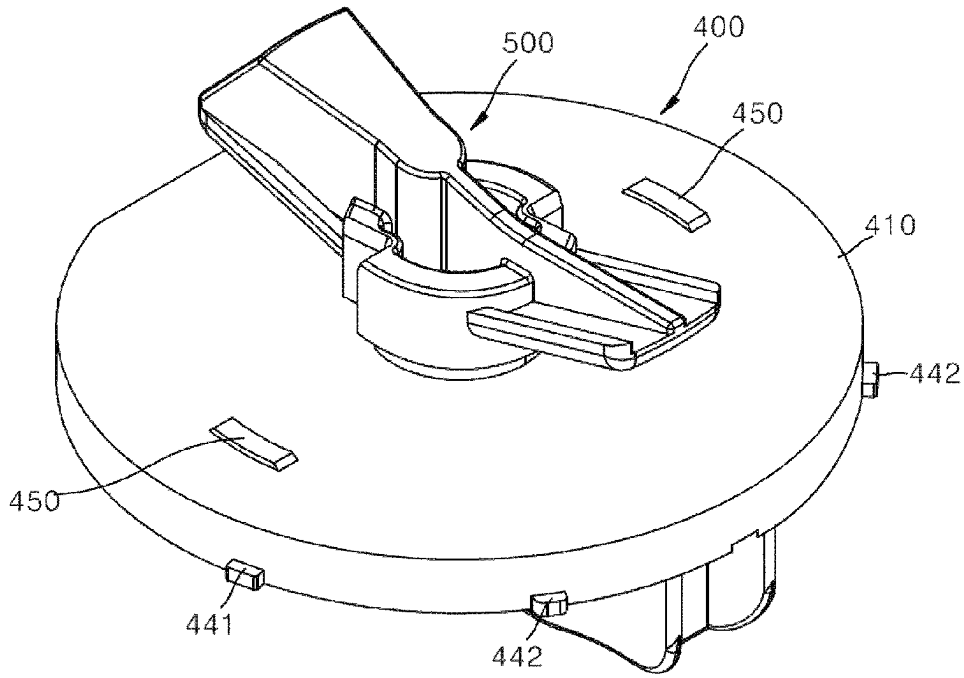


FIG. 17

(a1=a2)

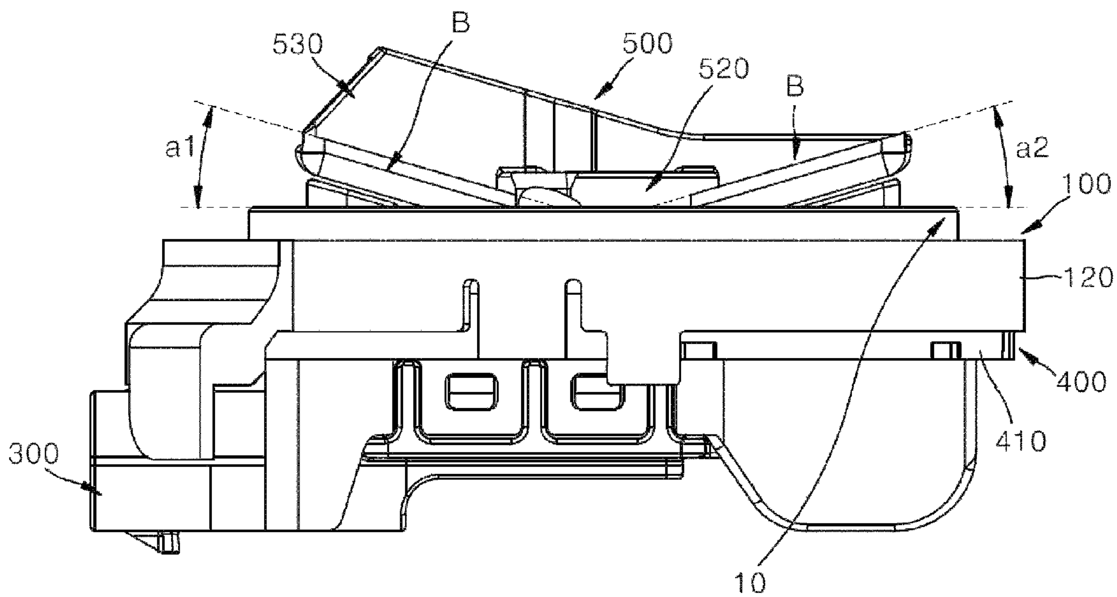


FIG. 18

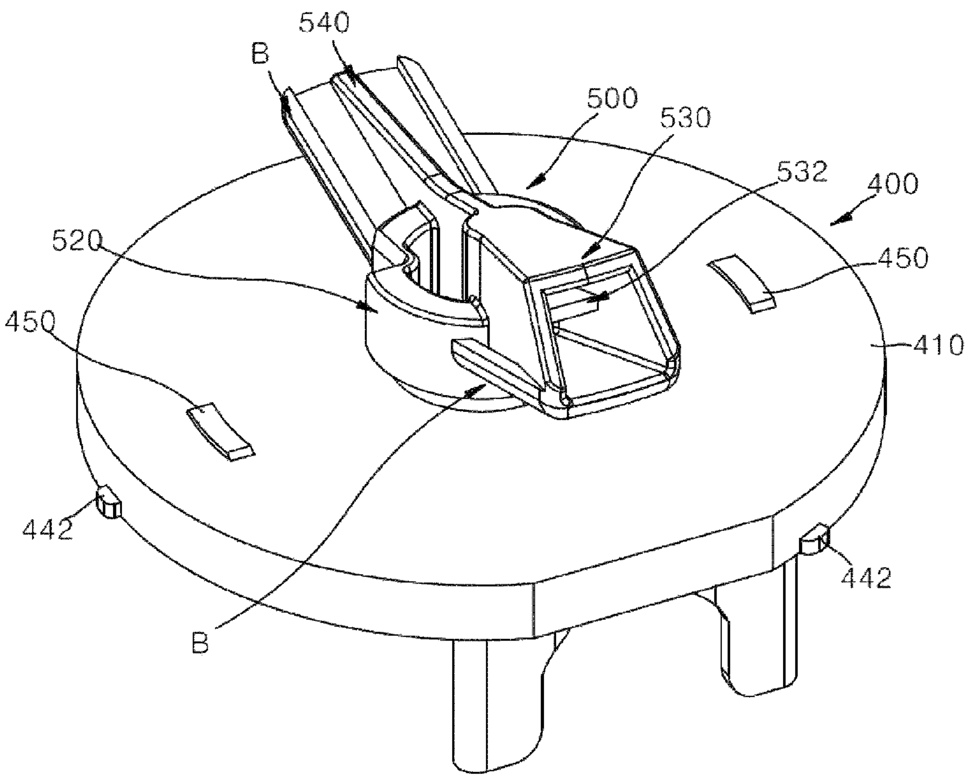
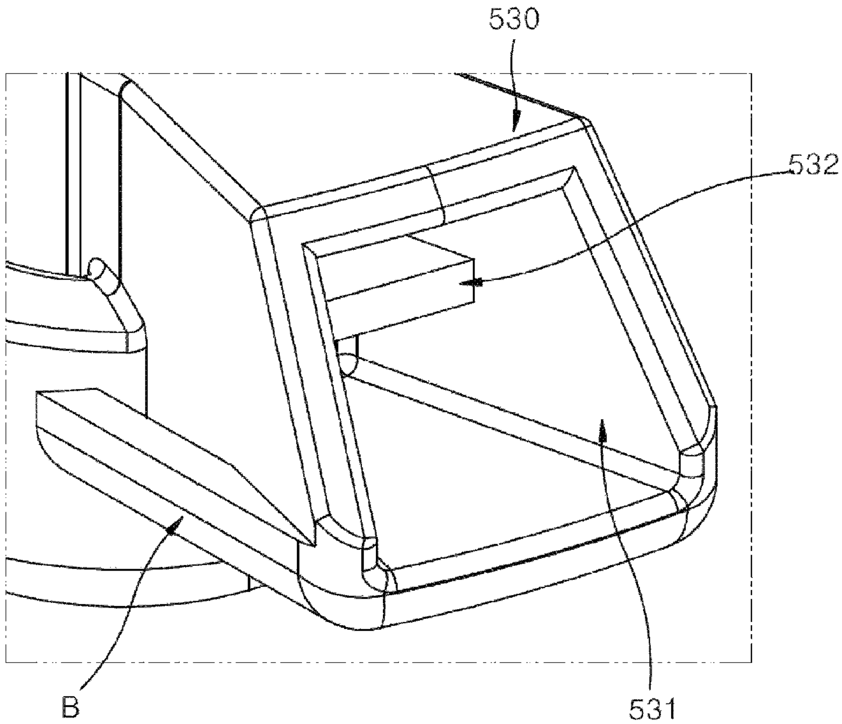


FIG. 19



FIXING DEVICE FOR INSTALLING PRESSURE-TYPE AIRBAG SENSOR

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a national stage filing under 35 U.S.C § 371 of PCT application number PCT/KR2021/007823 filed on Jun. 22, 2021, which is based upon and claims the benefit of priority to Korean Patent Application No. 10-2020-0077748 filed on Jun. 25, 2020, in the Korean Intellectual Property Office, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present invention relates to a fixing device for installing a pressure-type airbag sensor.

DESCRIPTION OF RELATED ART

Side airbags are installed in vehicles to protect drivers.

In the event of a side impact of the vehicle, an output signal from a side impact sensor installed on a side surface of the vehicle is input to an airbag control unit (ACU), and unfolding of the side airbag is controlled by the ACU.

In recent years, a pressure side impact sensor (PSIS) is mounted on a side portion of the vehicle to detect whether a side impact occurs.

The PSIS is mainly installed in a driver's door or a passenger's door. The PSIS serves to detect a pressure change generated inside a door momentarily due to deformation of the door in the event of the side impact and transmit the detected pressure change to the ACU.

Since an AK-LV 29 standard PSIS is mounted while rotating using one hand, convenience of installation is enhanced through a structure of two handles. This handle structure allows more moment to be transmitted during the rotation. Further, since the PSIS should not be loosened due to durability conditions or vibrations of the vehicle after the PSIS is completely installed, a loosening prevention part is provided.

In the PSIS according to the related art, a structure for assembling a plurality of components is complicated. For example, since structures such as a rotational direction fixing part that restricts movement in a rotational direction and the loosening prevention part that prevents separation between the components are additionally required, the structure is complicated.

The related document related to the present invention is JP-08040185.

RELATED ART DOCUMENT

Patent Document

(Patent Document 001) JP 08040185

SUMMARY OF THE DISCLOSURE

The present invention is directed to providing a fixing device for installing a pressure-type airbag sensor, in which a holder and a cover are prevented from being separated from each other, when the cover (**400**) passes through a section of the protrusion parts (**140**) and rotates to a set fastening position, one surface of the cover (**400**) is separated from the protrusion parts (**140**) and returns to its

original position, and in this case, the seal (**130**) is in elastic contact with the cover (**400**), and thus smooth sealing can be achieved at a position at which the rotation is completed.

Further, the present invention is directed to also providing a fixing device for installing a pressure-type airbag sensor, in which a connector housing (**300**) is covered through a prevention plate (**152**) before fastening between the holder (**100**) and the cover (**400**) is completed, and thus fastening of the connector housing (**300**) can be prevented from the outside.

The purposes of the present invention may be not limited to the purposes described above, and other purposes and advantages of the present invention that are not described may be understood by the following description and may be more clearly understood by embodiments of the present invention. Further, it may be easily identified that the purposes and advantages of the present invention may be implemented by units and combinations thereof described in the appended claims.

One aspect of the present invention provides a fixing device for installing a pressure-type airbag sensor.

The fixing device for installing a pressure-type airbag sensor includes a holder (**100**) having a disk shape, a cover (**400**) coupled to the holder (**100**) and having a sensor unit disposed at an upper end thereof, and a connector housing (**300**) disposed at a lower end of the cover (**400**).

The holder (**100**) may have a holder body (**110**) formed in a disc shape.

An outer wall (**120**) having a predetermined height may be formed on an outer circumference of the holder body (**110**).

A seal having a ring shape may be insert-injected into the holder body (**110**) to have a predetermined radius

Protrusion parts (**140**) having a predetermined curvature are formed on an upper surface and a lower surface of the holder body (**110**).

The protrusion parts (**140**) may be formed to protrude in a pair from each of the upper surface and the lower surface of the holder body (**110**).

The pair of protrusion parts (**140**) may be positioned to be symmetrical to each other with a central hole (**111**) formed at a center of the holder body (**110**) as a boundary on each of the upper surface and the lower surface of the holder body (**140**).

The pair of protrusion parts (**140**) formed on each of the upper surface and the lower surface of the holder body (**110**) may be arranged at the same position.

Locking parts (**180**) protruding upward may be formed to protrude from a plurality of positions on the outer circumference of the holder body (**110**).

The plurality of locking parts (**180**) may be formed.

The locking parts (**180**) may be formed in a shape bent in an L shape toward an inside of the holder body (**110**).

Each of the locking parts (**180**) may have a neck (**181**) extending from an outer wall thereof and a hook (**182**) bent inward from an upper end of the neck (**181**).

The plurality of locking parts (**180**) may be arranged at intervals of 60 degrees.

Rotation locking parts (**190**) may be formed at a plurality of positions on the outer circumference of the holder body (**110**).

Each of the rotation locking parts (**190**) may be formed in a rectangular plate shape.

Each of the rotation locking parts (**190**) may be formed by being cut in a rectangular shape from the outer circumference of the holder body (**110**).

A length of each of the rotation locking parts (**190**) may be formed longer than a thickness of the holder body (**110**).

The rotation locking parts (**190**) may be positioned on side portions of the locking parts (**180**), respectively.

A thickness of each of the rotation locking parts (**190**) may gradually become thicker from one side to the other side.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view illustrating a state in which a fixing device for installing an airbag sensor according to the present invention is disposed on a panel.

FIG. **2** is perspective views illustrating the fixing device for installing an airbag sensor according to the present invention.

FIG. **3** is a perspective view illustrating a state in which a cover and a holder according to the present invention are coupled.

FIG. **4** is a perspective view illustrating a configuration of the holder according to the present invention.

FIG. **5**A is a front view illustrating the holder according to the present invention.

FIG. **5**B is a perspective view illustrating a bottom surface of the holder according to the present invention.

FIG. **6** is a perspective view illustrating a state in which the cover and the holder according to the present invention are fastened.

FIG. **7** is a perspective view illustrating a bottom surface of the holder according to the present invention.

FIG. **8** are views illustrating a holder body in which a spring part is formed according to the present invention.

FIG. **9** is a perspective view illustrating a coupling relationship between the spring part and a sensor unit according to the present invention.

FIG. **10** is a plan view illustrating the holder in which a locking part is formed according to the present invention.

FIG. **11**A is an enlarged perspective view illustrating the locking part according to the present invention.

FIG. **11**B is an enlarged perspective view illustrating a rotation locking part according to the present invention.

FIG. **12** is a perspective view illustrating a bottom surface of the cover according to the present invention.

FIG. **13** are perspective views illustrating the bottom surface of the cover according to the present invention.

FIG. **14** are views illustrating separation prevention grooves according to the present invention.

FIG. **15** is a bottom view illustrating the cover according to the present invention.

FIG. **16** is perspective view illustrating the cover according to the present invention.

FIG. **17** is a side view illustrating an arrangement state of the sensor unit formed in the cover according to the present invention.

FIG. **18** is perspective view illustrating the cover according to the present invention.

FIG. **19** is a perspective view illustrating a pressure inlet of FIG. **18**.

DETAILED DESCRIPTION OF EXEMPLARY IMPLEMENTATIONS

Hereinafter, provision or arrangement of an arbitrary component on an "upper portion (or a lower portion)" of a substrate or "on (or under)" the substrate means that the arbitrary component is provided or arranged in contact with an upper surface (or a lower surface) of the substrate.

Further, the present invention is not limited to a state in which another component is not included between the substrate and the arbitrary component provided or arranged on (or under) the substrate.

Hereinafter, a fixing device for installing a pressure-type airbag sensor according to the present invention will be described with reference to the accompanying drawings.

The fixing device for installing a pressure-type airbag sensor according to the present invention includes a holder and a cover.

The holder according to the present invention will be described.

FIG. **1** is a perspective view illustrating a state in which a fixing device for installing a pressure-type airbag sensor according to the present invention is disposed on a panel, and FIG. **2** are perspective views illustrating the fixing device for installing a pressure-type airbag sensor according to the present invention.

The fixing device for installing a pressure-type airbag sensor according to the present invention is fixed to a panel **10**.

The fixing device for installing a pressure-type airbag sensor has a holder **100** and a cover **400**. A sensor unit **500** is disposed at an upper end of the cover **400**, and a connector housing **300** is disposed at a lower end of the cover **400**.

FIG. **3** is a perspective view illustrating a state in which a cover and a holder according to the present invention are coupled, and FIG. **4** is a perspective view illustrating a configuration of the holder according to the present invention.

Referring to FIGS. **3** and **4**, the holder **100** according to the present invention has a holder body **110** formed in a disc shape. An outer wall **120** having a predetermined height is formed on an outer circumference of the holder body **110**.

A seal **130** having a ring shape is insert-injected into the holder body **110** to have a predetermined radius. The seal **130** protrudes from an upper surface and a lower surface of the holder body **110** at a predetermined level.

A central hole **111** having an entirely rectangular shape is cut and formed in a center of the holder body **110**.

Further, protrusion parts **140** having a predetermined curvature are formed on the upper surface and the lower surface of the holder body **110** according to the present invention.

Each of the protrusion parts **140** forms a predetermined curvature.

The protrusion parts **140** are formed to protrude from each of the upper surface and the lower surface of the holder body **110** in a pair.

The pair of protrusion parts **140** are formed symmetric to each other with respect to the central hole **111** as boundary on each of the upper surface and the lower surface of the holder body **140**.

The pair of protrusion parts **140** formed on the upper surface and the lower surface of the holder body **110** are arranged at the same position.

A height of the protrusion parts **140** may be lower than a height of the seal **130**.

Further, an outer circumferential length of the protrusion parts **140** may be a quarter of the entire circumferential length connecting the protrusion parts **140**.

The holder body **110** according to the present invention is configured to rotate, and the seals **130** arranged on both surfaces of the holder body **110** form sealing due to contact with the cover **400** that is a counterpart.

In this case, while the cover **400** rotates, the cover **400** comes into physical contact with a section in which the protrusion parts **140** are formed and thus rotates while lifted by the height of the protrusion parts **140**. Accordingly, when the cover **400** rotates, while the seal **130** is protected from the cover **400** that is a counterpart, the holder body **110** may rotate.

Further, when the cover **400** passes through a section of the protrusion parts **140** and is rotated to a set fastening position, one surface of the cover **400** is separated from the protrusion parts **140** and returns to its original position. In this case, the seal **130** may be in elastic contact with the cover **400** to achieve smooth sealing at a position at which rotation is completed.

An upper end of the protrusion part **140** may be formed in a rounded shape.

FIG. **5**A is a front view illustrating the holder according to the present invention, and FIG. **5**B is a perspective view illustrating a bottom surface of the holder according to the present invention.

Referring to FIGS. **5**A and **5**B, a connector insertion prevention part **150** is formed on the outer circumference of the holder body **110** according to the present invention.

The connector insertion prevention part **150** has a rib **151** having a predetermined length and a prevention plate **152** protruding outward from an end of the rib **151**.

The rib **151** and the prevention plate **152** are formed integrally. A position of the prevention plate **152** changes while the holder body **110** rotates.

The prevention plate **152** may be formed in the holder body **110** to be positioned at a position covering the front side of the connector housing **300** in a position of the cover **400** before the rotation.

Further, when the holder body **110** is rotated to a position at which the holder body **110** is completely fastened to the cover **400**, the prevention plate **152** is positioned on a lateral side of the connector housing **300**. Accordingly, when the holder **100** and the cover **400** are completely fastened, the connector housing **300** disposed on the upper surface of the cover **400** forms an open state.

That is, the connector housing **300** is covered through the prevention plate **152** before fastening between the holder **100** and the cover **400** is completed, and thus fastening of the connector housing **300** can be prevented from the outside.

Further, the prevention plate **152** is formed to have a rounded curvature from one end to the other end. Therefore, as a slip is induced in the event of an impact with another external structure, damage of the holder **100** can be prevented.

FIG. **6** is a perspective view illustrating a state in which the cover and the holder according to the present invention are fastened, and FIG. **7** is a perspective view illustrating a bottom surface of the holder according to the present invention.

Referring to FIGS. **6** and **7**, first and second assembly guides **161** and **162** are formed at an upper end of the holder body **110** according to the present invention.

The assembly guides **161** and **162** have a pair of first assembly guides **161** protruding from the upper surface of the holder body **110** at both sides of one end of the central hole.

The pair of first assembly guides **161** form a predetermined distance therebetween. The pair of first assembly guides **161** may be caught by both sides of an inner circumference of one end of a fastening hole **11** formed in the panel **10**.

The assembly guides **161** and **162** have a pair of second assembly guides **162** protruding from the upper surface of the holder body **110** at both sides of the other end of the central hole **111**.

The pair of second assembly guides **162** are formed as plates having a cross section having a right triangle shape.

The pair of second assembly guides **162** form a predetermined distance therebetween. The pair of assembly guides **162** are supported by both sides of an inner circumference of the other end of the fastening hole **11** formed in the panel **10**.

In addition, rotation guides **163** having a predetermined curvature are formed on both sides of a central portion of the center hole **111** of the holder body **110** according to the present invention.

The rotation guides **163** may be configured as a pair of rotation guides **163** while forming a semicircular shape. The pair of rotation guides **163** are arranged to face each other.

The pair of rotation guides **163** are formed thicker than a thickness of the nearby holder body **110** and form a predetermined reinforcement force.

The pair of rotation guides **163** guide rotation of the sensor unit **500** formed on the upper surface of the cover **400**. A rotation guide member **510** having a circular shape is formed at a central portion of the sensor unit **500**.

Since rotation of the rotation guide member **510** is guided along an inner circumference of the pair of rotation guides **163** and the rotation guide member **510** has a step having a predetermined thickness, the rotation of the rotation guide member **510** of the sensor unit **500** during the rotation can be guided stably.

Although not illustrated in the drawings, a ring-shaped protrusion line is formed on the inner circumference of the pair of rotation guides **163**, the protrusion line rotates while fitted in a protrusion line groove formed in an inner circumference of the rotation guide member **510**, and thus more stable rotation can also be guided.

FIG. **8** are views illustrating a holder body in which a spring part is formed according to the present invention, and FIG. **9** is a perspective view illustrating a coupling relationship between the spring part and a sensor unit according to the present invention.

Referring to FIGS. **8** and **9**, a catching step **511** is formed on an outer circumference of the rotation guide member **510** of the sensor unit **500** according to the present invention.

Further, L-shaped cutout grooves **112** are formed on both sides of a central portion of the central hole **111** formed in the holder body **110**.

The spring parts **170** according to the present invention are formed in the cutout grooves **112**. Each of the spring parts **170** has a curvature rib **171** having a curvature and a catching rib **172** extending from a distal end of the curvature rib **171** in a straight line.

The end of the curvature rib **171** is formed integrally with an inner surface of the cutout groove **112**.

Accordingly, each of the spring parts **170** is formed in a free end shape.

In a state in which the cover **400** is coupled to the central hole **111**, a distal end of the catching rib **172** of the spring part **170** is exposed to the central hole **111** and thus may be caught by the catching step **511** of the rotation guide member **510** of the sensor unit **500**.

Here, the catching step **511** is formed in an L shape, and an end of the catching rib **172** is formed in a quadrangular shape.

Further, the catching rib **172** is caught by the upper surface of the cover **400**. Accordingly, it is possible to prevent the cover **400** from being separated upward.

Further, each of the spring parts **710** includes the curvature rib **171** and the catching rib **172** to form a predetermined length and thus has elasticity corresponding to the distance. Therefore, fastening sense between the holder **100** and the cover **400** can be improved, and assembly thereof can be secured.

In addition, the catching step **511** is formed in an L shape, and the end of the catching rib **172** caught thereby is formed in a quadrangular shape. Thus, the catching step **511** prevents the rotation before the cover **400** rotates and is completely fastened to the holder **100** and restricts reverse rotation of the cover **100** after the assembly.

FIG. **10** is a plan view illustrating the holder in which a locking part is formed according to the present invention, and FIG. **11**A is an enlarged perspective view illustrating the locking part according to the present invention.

Referring to FIGS. **10** and **11**A, locking parts **180** protruding upward are formed to protrude from a plurality of positions on the outer circumference of the holder body **110** according to the present invention.

The plurality of locking parts **180** are formed. The locking part **180** is formed in a shape bent in an L shape toward an inner side of the holder body **110**.

The locking part **180** has a neck **181** extending from an outer wall thereof and a hook **182** bent inward from an upper end of the neck **181**.

Further, the plurality of locking parts **180** may be arranged at intervals of 60 degrees.

Therefore, when the cover **400** is rotated to a fastening position while coupled to the holder **100**, ends of the locking parts **170** are caught by and fixed to an outer circumference of the lower end of the cover **400**. Therefore, it is possible to prevent the cover **400** from being separated from a lower portion of the holder **100**.

Further, although not illustrated in the drawings, insertion grooves into which the ends of the locking parts **180** are inserted may be further formed at a plurality of positions on the outer circumference of the lower end of the cover **400**.

FIG. **11**B is an enlarged perspective view illustrating a rotation locking part according to the present invention.

Referring to FIG. **11**B, rotation locking parts **190** are formed at the plurality of positions on the outer circumference of the holder body **110** according to the present invention.

Each of the rotation locking parts **190** is formed in a rectangular plate shape and is formed by being cut in a rectangular shape from the outer circumference of the holder body **110**.

Further, a length of each of the rotation locking parts **190** is formed longer than the thickness of the holder body **110**.

The rotation locking parts **190** are positioned on side portions of each of the locking parts **180**.

In addition, the thickness of each of the rotation locking parts **190** gradually becomes thicker from one side to the other side.

Therefore, when the cover **400** is rotated on an inner circumference of the holder body **110**, one end of each of the rotation locking parts **190** comes into contact with the outer circumference of the cover **400**, the cover **400** is rotated to the fastening position, and thus the outer circumference of the cover **400** is gradually pressed into contact with a thicker side thereof.

Therefore, the thickness of the rotation locking parts **190** may be gradually increased in a fastening direction, and when the cover **400** rotates, a rotational force may be gradually reduced. Further, a fastening sound is generated at this fastening position so that an operator can recognize the fastening.

Below, the cover according to the present invention will be described.

FIG. **12** is a perspective view illustrating a bottom surface of the cover according to the present invention.

Referring to FIG. **12**, the cover **400** according to the present invention has a cover body **410** formed in a disc shape.

A pair of first support portions **421** protruding upward are formed at one end of one surface of the cover body **410**. The pair of first support portions **421** protrude upward. The pair of first support portions **421** are formed to face each other and have an L-shaped cross section.

A second support portions **422** protruding upward is formed at the other end of one surface of the cover body **410**.

Here, the connector housing **300** is disposed in a space between the pair of first support portions **421** and the second support portion **422**.

Both sides of one end of the connector housing **300** are supported by the pair of first support portions **421**, and the other end of the connector housing **300** is supported by the second support portion **422**.

Further, the pair of first support portions **421** and the second support portion **422** protrude upward along the connector housing **300**.

Accordingly, the pair of first support portions **421** and the second support portion **422** according to the present invention may serve as a hand tool that may be held by the operator when the cover **100** is installed in the panel **10**.

Further, corners of the pair of first support portions **421** and the second support portion **422** are formed in a rounded shape, so that the operator can be protected and damage caused by a collision with an object can be prevented.

Further, the pair of first support portions **421** and the second support portion **422** may perform guiding when the connector housing **300** is assembled.

Further, when the cover **400** is installed in the panel **10**, the pair of first support portions **421** and the second support portion **422** can prevent the connector housing **300** from being caught by the panel **10**.

FIG. **13** are perspective views illustrating the bottom surface of the cover according to the present invention, and FIG. **14** are views illustrating separation prevention grooves according to the present invention.

Referring to FIGS. **13** and **14**, a seating part **430** in which an inner space is formed and a corner is formed in a wall body is formed between the pair of first support portions **421** and the second support portion **422** according to the present invention.

The connector housing **300** is seated on the inner space of the seating part **430**.

Here, cutout grooves **431** and **432** are formed at one end and the other end of the seating part **430**.

Further, a second insertion groove **422***a* is formed in the second support portion **422**, and the second insertion groove **422***a* is connected to the cutout groove **431** formed at the other end of the seating part **430**.

Further, a first insertion groove **421***a* is formed between the pair of first support portions **421**, and the first insertion groove **421***a* is connected to the cutout groove **432** formed at the one end of the seating part.

Here, the first and second insertion grooves **421***a* and **422***a* are coupled as fitting structures formed at both ends of the connector housing **300** are fitted in the first and second insertion grooves **421***a* and **422***a*. Therefore, the first and second insertion grooves **421***a* and **422***a* can guide the coupling of the connector housing **300** and prevent separation of the connector housing **300**.

Further, inclined surfaces S inclined inward are formed in entrances of the first and second insertion grooves **421***a* and **422***a*. The inclined surfaces S serve to guide the fitting structures of the connector housing **300** such that the fitting structures are inserted.

FIG. **15** is a bottom view illustrating the cover according to the present invention.

Referring to FIG. **15**, reverse rotation prevention protrusions **441** protruding outward are formed at intervals at a plurality of positions of an outer circumference of the cover body **410** according to the present invention. The reverse rotation prevention protrusions **441** are formed in a quadrangular shape.

Further, rotation guide protrusions **442** protrude from the plurality of positions of the cover body **410**. The rotation guide protrusions **422** may be formed in a quadrangular shape, and corners thereof have a predetermined curvature.

Further, the rotation guide protrusion **442** is formed to correspond to the number of rotation locking parts **190** formed on the outer wall of the holder body **110**.

Accordingly, when the holder body **110** and the cover body **410** are coupled and the cover body **410** rotates in an arrow direction, the rotation guide protrusions **442** move along a surface of which the thickness increases from one end to the other end of the rotation locking parts **190**, and the rotation guide protrusions **442** pass through the rotation locking parts **190** at distal ends. In this case, a fastening sound may be generated while the rotation guide protrusions **442** are separated from the distal ends of the rotation locking parts **190**.

Further, the reverse rotation prevention protrusions **441** are caught by the other ends of the rotation locking parts **190** having a relatively large thickness, so that the cover body **410** according to the present invention can be prevented from being reversely rotated in a direction opposite to the arrow direction.

FIG. **16** is perspective view illustrating the cover according to the present invention.

Referring to FIG. **16**, auxiliary protrusion parts **450** are formed at two positions on an upper surface of the cover body **410** according to the present invention. The auxiliary protrusion parts **450** may be formed as a pair or more of auxiliary protrusion parts **450**.

The auxiliary protrusion parts **450** may have substantially the same height as the protrusion parts **140** formed in the holder body **110**.

The seal **130** arranged on both sides of the holder body **110** in the present invention form sealing due to contact with a counterpart.

In this case, while the cover body **410** rotates, the cover body **400** comes into physical contact with a section in which the auxiliary protrusion parts **450** are formed and thus rotates while lifted by the height of the auxiliary protrusion parts **450**. Accordingly, when the cover body **410** rotates, the cover body **410** may rotate relative to the holder body **110** while protecting the seal **130**.

Further, when the holder body **110** passes through a section of the protrusion parts **140** and is rotated to a fastening position, one surface of the holder body **110** that is a counterpart is separated from the auxiliary protrusion parts **450** and returns to its original position. In this case, the seal **130** may be in elastic contact with the upper surface of the cover body **410** to achieve smooth sealing at a position at which the rotation is completed, and thus fastening sense can be improved, and a rotational force can be reduced during the rotation.

Here, an upper surface of each of the auxiliary protrusion parts **450** may also be formed in a convex curvature.

FIG. **17** is a side view illustrating an arrangement state of the sensor unit formed in the cover according to the present invention.

Referring to FIG. **17**, the sensor unit **500** is formed at a center of an upper end of the cover body **410** according to the present invention.

The sensor unit **500** has a connection body **520** formed at a center of the cover body **410**, a pressure inlet body **530** in which a pressure inlet **531** extending to one side of the connection body **520** is formed, and an extension body **540** extending to the other side of the connection body **520**.

Here, blades B are formed on both side surfaces of the pressure inlet body **530** and the extension body **540**.

Each of the blades B forms a predetermined inclination rising along an upper side from the center of the cover body **410**. The inclinations of the blades B are identical to each other (a**1**=a**2**). Therefore, the cover body **410** may advantageously be installed in panels having different thicknesses according to an inclination angle.

Further, the blades B according to the present invention are formed to protrude at regular intervals from the assembly guides formed on the holder body **110**. Therefore, when the cover body **410** is installed in the panel **10**, the blades B of the sensor unit **500** may be arranged to pass through the panel **10** and to be caught by an upper surface of the panel **10**.

FIG. **18** is perspective view illustrating the cover according to the present invention, and FIG. **19** is a perspective view illustrating a pressure inlet of FIG. **18**.

Referring to FIGS. **18** and **19**, in the pressure inlet body **530** according to the present invention, the pressure inlet **531** is formed.

A layered partition member **532** is formed inside the pressure inlet **531**. The partition member **532** is positioned inside an input end of the pressure inlet **531**.

Therefore, foreign substances may be filtered out by the partition member **532** formed in the pressure inlet **531**.

According to the configuration and operation described above, in the present invention, the holder and the cover are prevented from being separated from each other, and when the cover **400** passes through a section of the protrusion parts **140** and rotates to a set fastening position, one surface of the cover **400** is separated from the protrusion parts **140** and returns to its original position. In this case, the seal **130** is in elastic contact with the cover **400**, and thus smooth sealing can be achieved at a position at which the rotation is completed.

Further, in the present invention, the connector housing **300** is covered through the prevention plate **152** before fastening between the holder **100** and the cover **400** is completed, and thus fastening of the connector housing **300** can be prevented from the outside.

In the present invention, a holder and a cover are prevented from being separated from each other, and when the cover (**400**) passes through a section of the protrusion parts (**140**) and rotates to a set fastening position, one surface of the cover (**400**) is separated from the protrusion parts (**140**) and returns to its original position. In this case, the seal (**130**) is in elastic contact with the cover (**400**), and thus smooth sealing can be achieved at a position at which the rotation is completed.

Further, in the present invention, the connector housing (**300**) is covered through the prevention plate (**152**) before fastening between the holder (**100**) and the cover (**400**) is completed, and thus fastening of the connector housing (**300**) can be prevented from the outside.

Detailed descriptions of the present invention have been described above, but it is obvious that various modifications may be made without departing from the scope of the present invention.

Therefore, the scope of the present invention is not limited to the described embodiments and should be defined by equivalents of the appended claims as well as the scope of the appended claims.

That is, it should be understood that the above-described embodiments are illustrative in all aspects and not restrictive, the scope of the present invention is indicated by the appended claims described below rather than the detailed description, and it should be construed that the meaning and scope of the appended claims and all changes and modifications derived from equivalent concepts thereof are included in the scope of the present invention.

What is claimed is:

1. A fixing device for installing a pressure-type airbag sensor, the fixing device comprising:
- a holder having a disk shape;
- a cover coupled to the holder and having a sensor unit disposed at an upper end thereof; and
- a connector housing disposed at a lower end of the cover,
- wherein the holder comprises:
  - a holder body formed in a disc shape,
  - an outer wall having a predetermined height and formed on an outer circumference of the holder body,
  - a seal having a ring shape is insert-injected into the holder body to have a predetermined radius, and
  - protrusion parts having a predetermined curvature are formed on an upper surface and a lower surface of the holder body,
  - wherein
    - a height of the protrusion parts is lower than a height of the seal,
    - the holder body is configured to rotate relative to the cover, and
    - the seal forms sealing between the holder body and the cover, and
- wherein
  - while the cover rotates, one surface of the cover comes into physical contact with a section in which the protrusion parts are formed and lifted by the height of the protrusion parts, and
  - when the cover passes through the section of the protrusion parts and is rotated to a set fastening position, the one surface of the cover is separated from the protrusion parts and returns to its original position.

2. The fixing device of claim **1**, wherein the protrusion parts are formed to protrude in a pair from each of the upper surface and the lower surface of the holder body,
- the pair of protrusion parts are positioned to be symmetrical to each other with a central hole formed at a center of the holder body as a boundary on each of the upper surface and the lower surface of the holder body, and
- the pair of protrusion parts formed on each of the upper surface and the lower surface of the holder body are arranged at the same position.

3. The fixing device of claim **1**, wherein locking parts protruding upward are formed to protrude from a plurality of positions on the outer circumference of the holder body,
- the locking parts are formed in a shape bent in an L shape toward an inside of the holder body,
- each of the locking parts has a neck extending from an outer wall thereof and a hook bent inward from an upper end of the neck, and
- the plurality of locking parts are arranged at intervals of 60 degrees.

4. The fixing device of claim **3**, wherein rotation locking parts are formed at a plurality of positions on the outer circumference of the holder body,
- each of the rotation locking parts is formed in a rectangular plate shape and is formed by being cut in a rectangular shape from the outer circumference of the holder body,
- a length of each of the rotation locking parts is formed longer than a thickness of the holder body, and
- the rotation locking parts are positioned on side portions of the locking parts, respectively.

5. The fixing device of claim **4**, wherein a thickness of each of the rotation locking parts gradually becomes thicker from one side to the other side.

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